

VUI
FALL/SPRING MID/FINAL TERM 2050 (Year) Date: 22/12/2050
Campus Code: Pakis fan Course Code: 8502 Student ID: ma nahi btao

Rough Sheet
Superintendent Signature: [Signature]
Student Signature: [Signature]

Define Short path and find 6 to 6 vertex

Draw Prim algorithm start from D

Draw adjacency matrix

	1	2	3	4
1	0	1	1	1
2	1	0	0	0
3	1	0	0	1
4	1	0	1	0

up in $m(i, i, n)$.

Example: Let us go through an example. We want to find the optimal multiplication order for

$$A_1 \cdot A_2 \cdot A_3 \cdot A_4 \cdot A_5$$

$$20 \times 10 \times 40 \times 30 \times 20$$

We will compute the entries of the m matrix starting with the base condition. We first fill that main diagonal:

0				
	0			
		0		
			0	
				0

matris diya tha ya
table ma kol value nhi
thi seri fill karani
thi (LONG)

Next, we compute the entries in the first super diagonal, i.e., the diagonal above the main diagonal:

Ya LONG

Neachy Shorts

♥: What is the native evaluation 0/1 knapsack problem. How we can avoid exponential running Time using dynamic programming

The problem is called a "0/1" problem, because each item must be entirely accepted or rejected. How do we solve the problem. We could try the brute-force solution:

- Since there are n items, there are 2^n possible combinations of the items (an item either chosen or not).
- We go through all combinations and find the one with the most total value and with total weight less or equal to W .

♥: Coin change problems can be solved using dynamic programming and greedy Algorithm. Mentioned there running Time Both of greedy and dynamic programming

♥: Choose Yes or No and briefly explain it

If Problem A Reduce (polynomial problem)

A is NP-complete and B is NP-Complete

♥: The generic graph traversal a set of candidate edges put data structure called a bag. **Name of important property and give reasons.**

8.1.3 Generic Graph Traversal Algorithm

The generic graph traversal algorithm stores a set of candidate edges in some data structures we'll call a "bag". The only important properties of the "bag" are that we can put stuff into it and then later take stuff

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CHAPTER 8. GRAPHS

back out. Here is the generic traversal algorithm.

```

TRaverse(s)
1  put (s, s) in bag
2  while bag not empty
3  do take (p, v) from bag
4  if (v is unmarked)
5  then mark v
6  parent(v) ← p
7  for each edge (v, w)
8  do put (v, w) in bag
    
```

Short Question Bag! Important property vels

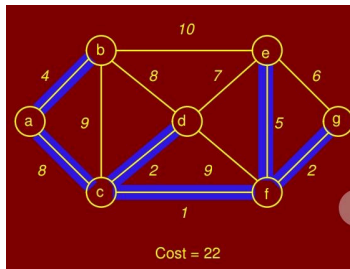
Notice that we are keeping edges in the bag instead of vertices. This is because we want to remember, whenever we visit v for the first time, which previously-visited vertex p put v into the bag. The vertex p is called the *parent* of v .

The running time of the traversal algorithm depends on how the graph is represented and what data structure is used for the bag. But we can make a few general observations.

- Since each vertex is visited at most once, the for loop in line 7 is executed at most V times.
- Each edge is put into the bag exactly twice: once as (u, v) and once as (v, u) , so line 8 is executed at most $2E$ times.
- Finally, since we can't take out more things out of the bag than we put in, line 3 is executed at most $2E + 1$ times.
- Assume that the graph is represented by an adjacency list so the overhead of the for loop in line 7 is constant per edge.

If we implement the bag by using a stack, we have *depth-first search* (DFS) or traversal.

- ♥: Dijkstra Algorithm is based on Greedy Approach. Prove it
- ♥: Cost of MST Diagram di thi



22 cost

MCQS

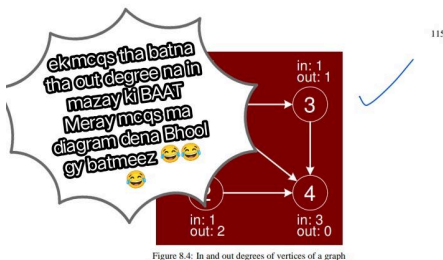
8.6.3 Bellman-Ford Algorithm

2 MCQS Jo highlight hain

Dijkstra's single-source shortest path algorithm works if all edges weights are non-negative and there are no negative cost cycles. Bellman-Ford allows negative weights edges and no negative cost cycles. The algorithm is slower than Dijkstra's, running in $O(V^2E)$ time.

Like Dijkstra's algorithm, Bellman-Ford is based on performing repeated relaxations. Bellman-Ford applies relaxation to every edge of the graph and repeats this $V - 1$ times. Here is the algorithm; it is

In a digraph, the number of edges coming out of a vertex is called the *out-degree* of that vertex. Number of edges coming in is the *in-degree*. In an undirected graph, we just talk of degree of a vertex. It is the number of edges incident on the vertex.



Consider the string "abacdaacac". If the string is coded with ASCII codes, the message length would be $10 \times 8 = 80$ bits. We will see shortly that the same string encoded with a variable length Huffman encoding scheme will produce a shorter message.

ek mees tha
je ma ?
character thy
bit batan thi
Jo 72 bit banti

17/101

CHAPTER 7. GREEDY ALGORITHMS

7.2.1 Huffman Encoding Algorithm

Here is how the Huffman encoding algorithm works. Given a message string, determine the frequency of occurrence (relative probability) of each character in the message. This can be done by parsing the message and counting how many times each character (or symbol) appears. The probability is the number of occurrence of a character divided by the total characters in the message. The frequencies and probabilities for the example string "abacdaacac" are

Mees the batan
Huffman
encoding ki
frequency phaly
kider Sy batan thi
largest Sy hoi

character	a	b	c	d
frequency	5	1	3	1
probability	0.5	0.1	0.3	0.1

7.1.2 Complexity of Coin Change Algorithm

Greedy algorithm (non-optimal) takes $O(k)$ time. Dynamic Programming takes $O(kN)$ time. Note that N can be as large as 2^k so the dynamic programming algorithm is really exponential in k .



Item i	Weight w_i	Value v_i
1	2	3
2	3	4
3	4	5
4	5	8
5	9	10

Figure 6.3: Knapsack can hold $W=20$

The knapsack problem belongs to the domain of optimization problems. Mathematically, the problem is

Item 2 and 3 select Karni thi weight 50 tha

If there are $\Theta(n^2)$ entries in edit distance matrix then the total running time is

Answer (Please select your correct option)

☐ $\Theta(1)$

☐ $\Theta(n^2)$ correct

☐ $\Theta(n)$

☐ $\Theta(n \log n)$

Using ASCII standard the string "ab-@!c:" will be encoded _____ bytes

Answer (Please select your correct option)

☐ 16

☐ 7

☐ 6

☐ This string cannot be stored using ASCII standard. correct Made b

Question No: 28 of 52

The greedy part of the Huffman encoding algorithm is to first find two nodes with _____ frequency.

Answer (Please select your correct option)

☐ Larger

☐ Smallest correct

☐ Balance

☐ Character Made b

Adding any edge to a free tree

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Answer (Please select your correct option)

☐ keeps it the free tree and increases the size of the tree.

☐ creates a unique cycle correct

☐ it is not allowed to add the edge in free tree

☐ creates multiple cycles. Made b

Kruskal Ford algorithm applies relaxation to every

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Answer (Please select your correct option)

☐ edge of the graph and repeats exactly $n-1$ times correct

☐ vertices of the graph and repeats exactly $E-1$ times

☐ edge of the graph and repeats exactly $E-1$ times

☐ edge but use the back edges for the completion Made b

Complexity when the comparison based merge and quick sort algorithms fall in

Answer (Please select your correct option)

☐ Deterministic Polynomial class

☐ Non-Deterministic Polynomial class

☐ Quick sort in P class and Merge sort in NP class

☐ Quick sort in NP class and Merge sort in P class Made b

Question No : 22 of 52

If we can solve a single NP problem in P time

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Answer (Please select your correct option) Vu

☐

All NP-problems can be solved

correct

☐

All P problems can be solved

☐

We cannot predict about the solution of other NP problems

☐

This can be never possible to solve the NP problem in P time

Made b

In the 3-coloring problem, for two vertices to be in the same group, they must be not _____ to each other.

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Answer (Please select your correct option) Vu

☐

Apart from

☐

Far from

☐

Near to

☐

Adjacent to

correct

Made b

Precedence constraint graph is

Answer (Please select your correct option) Vu

☐

non acyclic directed graph

☐

acyclic undirected graph

☐

non acyclic undirected graph

☐

acyclic directed graph

correct

Made b

In Prim's algorithm, the additional information maintained by the algorithm is:

Answer (Please select your correct option) Vu

☐

the length of the shortest path from vertex v to the vertex u

correct

☐

the length of the shortest edge from vertex v to points already in the tree

☐

the dynamic programming rules

☐

the information about all adjacent vertices

Made b

Question No : 26 of 52

In strongly connected components the component digraph is

Answer (Please select your correct option) Vu

☐

necessarily cyclic

correct

☐

necessarily acyclic

☐

not necessary it can be both cyclic and acyclic

☐

cyclic with some other constraints

Made b

Knuth's Algorithm has time complexity:

Answer (Please select your correct option) VU

☐ overall $O(V \log E)$

☐ overall $O(E \log V)$ for sparse graph $O(V \log E)$

☐ overall $O(E \log E)$ and for sparse graph $O(E \log V)$

☒ overall $O(E^2)$ and for sparse graph $O(V^2)$ correct

Made b

Question No: 22 of 52

In generic graph traversal algorithm we

Answer (Please select your correct option) VU

☐ put vertices in the bag data structure

☒ put edges in the bag data structure

☐ put edges in stack data structure

☐ put vertices in the stack data structure

Made

Bellman-Ford algorithm applies relaxation to every

Bellman-Ford applies relaxation to every edge of the graph and repeats this

Answer (Please select your correct option) VU

☐ edge of the graph and repeats exactly $v-1$ times correct

☐ vertex of the graph and repeats exactly $E-1$ times

☐ edge of the graph and repeats exactly $E-1$ times

☐ edge but use the back edges for the completion

Made b

Question No: 31 of 52

In NP-problems "NP" represents

The term "NP" does not mean "not polynomial". Originally, the term n

Answer (Please select your correct option) VU

☒ Non-deterministic Polynomial correct

☐ Null polynomial

☐ Negative Polynomial

☐ Non-polynomial

Made b

Question No: 32 of 52

The recurrence represented by $T(n) = \sum_{i=1}^n \sum_{j=1}^i T(j)$ has time complexity belongs to:

Answer (Please select your correct option) VU

☐ P-Class

☐ NP-Class

☐ Co-NP Class

☐ Unpredictable class

Made b

Question No: 36 of 52

If a problem "Q" is NP-complete it must be

Answer (Please select your correct option) VU

☒ NP and NP-hard

☐ NP not necessarily NP-Hard

☐ NP-Hard means it is NP complete as well

☐ in P and NP

Made